

The PWDG–UWVF Equivalence Revisited

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Abstract

The ultra-weak variational formulation (UWVF) is classically recovered from the plane-wave discontinuous Galerkin (PWDG) flux family for the standard one-half choice of the three flux parameters. We write this equivalence directly at the variational level by introducing outgoing and incoming Cessenat–Després impedance variables on each element boundary. In these variables, the PWDG form is a fixed scalar multiple of the UWVF form pulled back to broken Trefftz fields, so no intermediate residual variable is required. The two oriented impedance mismatches reproduce exactly the Dirichlet and normal-flux jump contributions of PWDG. The same calculation gives an impedance interpretation of arbitrary positive PWDG penalties and distinguishes the one-way UWVF family from a two-orientation forward–reverse coupling. Finally, the Trefftz-DG stability identity yields Riesz-normalized operator forms for the same discrete problem in broken-field and impedance-trace coordinates. An eight-element plane-wave calculation verifies the matrix equivalence to roundoff using analytic edge integration.

Keywords: Trefftz methods, plane-wave discontinuous Galerkin, ultra-weak variational formulation, impedance traces, Riesz map, conditioning

1. Introduction

The ultra-weak variational formulation (UWVF) of Cessenat and Després uses element-boundary impedance variables as the global unknowns [1, 2]. Plane-wave discontinuous Galerkin (PWDG), or more generally Trefftz-DG, uses broken Helmholtz fields and couples neighboring elements by numerical fluxes. The two descriptions have long been known to coincide for

$$\alpha = \beta = \delta = \frac{1}{2}; \tag{1}$$

see [3, Remark 2.1] and [7, Sec. 2.2].

The usual statement of the equivalence is often given by comparing numerical fluxes. Here we instead write the complete PWDG sesquilinear form as a UWVF. This has two advantages. First, the normalization is explicit: when the standard, unscaled impedance traces $-\partial_n + ik$ and $\partial_n + ik$ are used, PWDG is $i/(2k)$ times the UWVF in broken-field coordinates. Second, the two PWDG jump penalties can be read directly from the two orientations of the impedance transmission condition. The factors $1/2$ in (1) are then consequences of the impedance algebra rather than parameters that happen to reproduce an existing method.

The same calculation also clarifies general positive α and β . Their ratio fixes an effective Robin impedance,

$$\eta_{\alpha,\beta} = k\sqrt{\alpha/\beta},$$

whereas their product fixes the normalization and the balance between the two orientations of the impedance coupling. In particular, $4\alpha\beta = 1$ is the one-way UWVF curve, and the physical impedance $\eta_{\alpha,\beta} = k$ selects the symmetric point $\alpha = \beta = 1/2$. Away from that curve, the exact representation below is an affine combination of the forward and reverse orientations; when $4\alpha\beta < 1$, the reverse coefficient is negative.

A second theme is the Riesz geometry of the discrete problem. For Trefftz fields,

$$\operatorname{Im} A_h(v, v) = \|v\|_{\text{TDG}}^2. \quad (2)$$

Hence the skew-Hermitian part of the PWDG matrix is exactly i times the TDG Gram matrix. Its inverse square root is therefore the intrinsic Riesz change of coordinates, not an external algebraic scaling. The PWDG matrix becomes $K + iI$; the corresponding UWVF matrix is the same normalized operator up to the fixed scalar induced by the impedance normalization.

Throughout we use the impedance convention

$$\partial_n u + iku = g, \quad (3)$$

and the Cessenat–Després labeling in which $-\partial_n + ik$ is the outgoing trace and $\partial_n + ik$ is the incoming trace. Changing the time-harmonic sign convention conjugates the impedance traces and reverses the corresponding signs of i .

2. PWDG in broken Trefftz fields

Let $k > 0$ be real. Let $\Omega \subset \mathbb{R}^2$ be polygonal and let $\mathcal{T}_h$ be a conforming mesh with interior edges $\mathcal{F}_h^I$ and boundary $\Gamma = \partial\Omega$. We consider

$$-\Delta u - k^2 u = 0 \quad \text{in } \Omega, \quad \partial_n u + iku = g \quad \text{on } \Gamma. \quad (4)$$

Let V_h be a finite-dimensional broken Trefftz space, so every $v_h \in V_h$ satisfies $-\Delta v_h - k^2 v_h = 0$ on each element.

For an interior edge $e = \partial K \cap \partial K'$, with outward normals n_K and $n_{K'}$, define the actual DG jumps

$$[[v]]_N = v_K n_K + v_{K'} n_{K'}, \quad [[\nabla v]]_N = \partial_{n_K} v_K + \partial_{n_{K'}} v_{K'}, \quad (5)$$

and use the arithmetic averages $\{\{v\}\}$ and $\{\{\nabla v\}\}$.

The primal Trefftz-DG formulation [3, 7] is: find $u_h \in V_h$ such that

$$A_h(u_h, v_h) = \ell_h(v_h) \quad \forall v_h \in V_h, \quad (6)$$

where, for $\alpha, \beta > 0$ and $0 < \delta < 1$,

$$\begin{aligned} A_h(u, v) = & \int_{\mathcal{F}_h^I} \left(\{\{u\}\} [[\nabla \bar{v}]]_N - \{\{\nabla u\}\} \cdot [[\bar{v}]]_N + \alpha i k [[u]]_N \cdot [[\bar{v}]]_N \right. \\ & \left. - \beta (ik)^{-1} [[\nabla u]]_N [[\nabla \bar{v}]]_N \right) ds \\ & + \int_{\Gamma} \left((1 - \delta) i k u \bar{v} + (1 - \delta) u \overline{\partial_n v} - \delta \partial_n u \bar{v} - \delta (ik)^{-1} \partial_n u \overline{\partial_n v} \right) ds, \quad (7) \end{aligned}$$

and

$$\ell_h(v) = \int_{\Gamma} g \left((1 - \delta) \bar{v} - \delta (ik)^{-1} \overline{\partial_n v} \right) ds. \quad (8)$$

The associated TDG norm is

$$\begin{aligned} \|v\|_{\text{TDG}}^2 &= k^{-1} \|\beta^{1/2} \llbracket \nabla v \rrbracket_N\|_{\mathcal{F}_h^I}^2 + k \|\alpha^{1/2} \llbracket v \rrbracket_N\|_{\mathcal{F}_h^I}^2 \\ &\quad + k^{-1} \|\delta^{1/2} \partial_n v\|_{\Gamma}^2 + k \|(1 - \delta)^{1/2} v\|_{\Gamma}^2. \end{aligned} \quad (9)$$

3. PWDG written directly as a UWVF

For each element K , define the outgoing and incoming impedance variables

$$x_K(v) := -\partial_{n_K} v + ikv, \quad y_K(v) := \partial_{n_K} v + ikv. \quad (10)$$

Following Cessenat and Després, $x_K(v)$ is the outgoing impedance and $y_K(v)$ is the incoming impedance. The outgoing trace map is injective on every local Trefftz space. Indeed, if $x_K(v) = 0$, then $\partial_{n_K} v = ikv$ on ∂K , and Green's identity gives

$$\int_K (|\nabla v|^2 - k^2 |v|^2) dx = ik \int_{\partial K} |v|^2 ds.$$

The left-hand side is real, so $v = 0$ on ∂K ; then also $\partial_{n_K} v = 0$, and uniqueness for the Cauchy problem implies $v \equiv 0$ in K . Consequently the local scattering operator is well defined on the range of the outgoing trace and satisfies

$$y_K(v) = F_K x_K(v). \quad (11)$$

Thus the discrete UWVF trace space is precisely

$$\mathcal{X}_h^- := \{(x_K(v_h|_K))_{K \in \mathcal{T}_h} : v_h \in V_h\}.$$

If $\{\Phi_j\}$ is a basis of V_h , we use $\{(x_K(\Phi_j|_K))_K\}$ as the corresponding UWVF trace basis. With this identification, PWDG and UWVF matrices may be compared in the same coefficient vector without any hidden change of basis.

For $u, v \in V_h$, define the pulled-back standard UWVF form

$$\begin{aligned} \mathcal{U}_h(u, v) &= \sum_{K \in \mathcal{T}_h} \int_{\partial K} x_K(u_K) \overline{x_K(v_K)} ds \\ &\quad - \sum_{e = \partial K \cap \partial K' \in \mathcal{F}_h^I} \int_e \left(x_{K'}(u_{K'}) \overline{y_K(v_K)} + x_K(u_K) \overline{y_{K'}(v_{K'})} \right) ds, \end{aligned} \quad (12)$$

and

$$\mathcal{L}_h(v) = \sum_{K \in \mathcal{T}_h} \int_{\partial K \cap \Gamma} g \overline{y_K(v_K)} ds. \quad (13)$$

Equation (12) is exactly the usual UWVF with outgoing trace unknown x_K and incoming trace $y_K = F_K x_K$, pulled back through (10) to broken Trefftz fields.

Theorem 1 (PWDG is the pulled-back UWVF). *At*

$$\alpha = \beta = \delta = \frac{1}{2},$$

the PWDG formulation (6) is precisely

$$\frac{i}{2k} \mathcal{U}_h(u_h, v_h) = \frac{i}{2k} \mathcal{L}_h(v_h) \quad \forall v_h \in V_h. \quad (14)$$

Equivalently,

$$A_h(u, v) = \frac{i}{2k} \mathcal{U}_h(u, v), \quad \ell_h(v) = \frac{i}{2k} \mathcal{L}_h(v). \quad (15)$$

Thus the PWDG broken-field equation is a UWVF written in broken Trefftz coordinates.

Proof. Consider first one interior edge $e = \partial K \cap \partial K'$. The contribution of this edge to $\mathcal{U}_h$ is

$$\begin{aligned} \mathcal{U}_e(u, v) &= \langle x_K(u_K), x_K(v_K) \rangle_e + \langle x_{K'}(u_{K'}), x_{K'}(v_{K'}) \rangle_e \\ &\quad - \langle x_{K'}(u_{K'}), y_K(v_K) \rangle_e - \langle x_K(u_K), y_{K'}(v_{K'}) \rangle_e. \end{aligned} \quad (16)$$

Expanding the four complete impedance products and using $n_{K'} = -n_K$ gives

$$\begin{aligned} \frac{i}{2k} \mathcal{U}_e(u, v) &= \int_e \left(\{\{u\}\} \llbracket \nabla \bar{v} \rrbracket_N - \{\{\nabla u\}\} \cdot \llbracket \bar{v} \rrbracket_N \right) ds \\ &\quad + \frac{ik}{2} \langle \llbracket u \rrbracket_N, \llbracket v \rrbracket_N \rangle_e + \frac{i}{2k} \langle \llbracket \nabla u \rrbracket_N, \llbracket \nabla v \rrbracket_N \rangle_e. \end{aligned} \quad (17)$$

This is exactly the interior PWDG edge form with $\alpha = \beta = 1/2$.

On a physical boundary edge, the diagonal UWVF term gives

$$\frac{i}{2k} \langle x(u), x(v) \rangle_\Gamma = \int_\Gamma \left(\frac{ik}{2} u \bar{v} + \frac{1}{2} u \overline{\partial_n v} - \frac{1}{2} \partial_n u \bar{v} + \frac{i}{2k} \partial_n u \overline{\partial_n v} \right) ds, \quad (18)$$

which is the boundary part of (7) at $\delta = 1/2$. Finally,

$$\frac{i}{2k} \int_\Gamma g \overline{y(v)} ds = \int_\Gamma g \left(\frac{1}{2} \bar{v} + \frac{i}{2k} \overline{\partial_n v} \right) ds = \ell_h(v), \quad (19)$$

again because $-(ik)^{-1} = i/k$. Summing over all edges proves (15). $\square$

Remark 1 (Where the scalar factor comes from). *The factor $i/(2k)$ is a normalization, not a discrepancy between the methods. If the impedance variables are rescaled by $(2k)^{-1/2}$, then the pulled-back UWVF matrix is simply $-iA_P$. If the standard unscaled impedance variables (10) are retained, then the matrix relation is instead*

$$A_U^{\text{std}} = -2ik A_P. \quad (20)$$

The two versions have identical condition numbers and identical normalized spectra.

4. The impedance identity behind the PWDG jumps

The preceding theorem uses the physical impedance and the one-half parameters. We now keep $\alpha, \beta > 0$ arbitrary and show directly how the PWDG jumps are encoded by two oriented impedance transmission conditions.

Set

$$\eta_{\alpha,\beta} = k\sqrt{\alpha/\beta}, \quad \rho_{\alpha,\beta} = 2\sqrt{\alpha\beta}. \quad (21)$$

Define the corresponding weighted impedance variables

$$x_K^{\alpha,\beta}(v) := -\sqrt{\frac{\beta}{k}} \partial_{n_K} v + i\sqrt{\alpha k} v, \quad y_K^{\alpha,\beta}(v) := \sqrt{\frac{\beta}{k}} \partial_{n_K} v + i\sqrt{\alpha k} v. \quad (22)$$

Equivalently,

$$x_K^{\alpha,\beta}(v) = \sqrt{\beta/k} (-\partial_{n_K} + i\eta_{\alpha,\beta})v, \quad y_K^{\alpha,\beta}(v) = \sqrt{\beta/k} (\partial_{n_K} + i\eta_{\alpha,\beta})v.$$

Proposition 1 (Impedance variables and DG jumps). *Let $e = \partial K \cap \partial K'$ be an interior edge. Then*

$$\begin{aligned} & \frac{1}{2} \left\langle x_K^{\alpha,\beta}(u_K) - y_{K'}^{\alpha,\beta}(u_{K'}), x_K^{\alpha,\beta}(v_K) - y_{K'}^{\alpha,\beta}(v_{K'}) \right\rangle_e \\ & + \frac{1}{2} \left\langle x_{K'}^{\alpha,\beta}(u_{K'}) - y_K^{\alpha,\beta}(u_K), x_{K'}^{\alpha,\beta}(v_{K'}) - y_K^{\alpha,\beta}(v_K) \right\rangle_e \\ & = \alpha k \langle \llbracket u \rrbracket_N, \llbracket v \rrbracket_N \rangle_e + \frac{\beta}{k} \langle \llbracket \nabla u \rrbracket_N, \llbracket \nabla v \rrbracket_N \rangle_e. \end{aligned} \quad (23)$$

Thus the two oriented impedance mismatches on the left are exactly the complex packaging of the Dirichlet and normal-flux jumps on the right.

Proof. Because $n_{K'} = -n_K$,

$$x_K^{\alpha,\beta}(u_K) - y_{K'}^{\alpha,\beta}(u_{K'}) = -\sqrt{\frac{\beta}{k}} \llbracket \nabla u \rrbracket_N + i\sqrt{\alpha k} (\llbracket u \rrbracket_N \cdot n_K), \quad (24)$$

$$x_{K'}^{\alpha,\beta}(u_{K'}) - y_K^{\alpha,\beta}(u_K) = -\sqrt{\frac{\beta}{k}} \llbracket \nabla u \rrbracket_N - i\sqrt{\alpha k} (\llbracket u \rrbracket_N \cdot n_K). \quad (25)$$

The same identities hold for v . Adding the two impedance products cancels the mixed value-flux terms. Since $\llbracket u \rrbracket_N$ is parallel to n_K on e , the remaining terms are precisely the two jump products in (23). $\square$

At $\alpha = \beta = 1/2$, the weighted variables are exactly $(2k)^{-1/2}x_K$ and $(2k)^{-1/2}y_K$. Proposition 1 therefore explains the two interior one-half factors in the classical equivalence: the Dirichlet and normal-flux jump terms are the two-channel real representation of the two oriented UWVF impedance transmission conditions.

5. General positive PWDG penalties as weighted UWVF couplings

The previous proposition concerns the Riesz part of the edge form. The entire PWDG edge operator has an equally direct representation.

For an interior edge define the consistency and jump forms

$$C_e(u, v) = \int_e \left(\{u\} \llbracket \nabla \bar{v} \rrbracket_N - \{\nabla u\} \cdot \llbracket \bar{v} \rrbracket_N \right) ds, \quad (26)$$

$$G_e^{\alpha, \beta}(u, v) = \alpha k \langle \llbracket u \rrbracket_N, \llbracket v \rrbracket_N \rangle_e + \frac{\beta}{k} \langle \llbracket \nabla u \rrbracket_N, \llbracket \nabla v \rrbracket_N \rangle_e. \quad (27)$$

Then

$$A_{P,e}(u, v) = C_e(u, v) + iG_e^{\alpha, \beta}(u, v). \quad (28)$$

Using the weighted impedance variables (22), define the one-way pulled-back UWVF edge form by

$$\begin{aligned} \mathcal{U}_e^{\rightarrow}(u, v) = & \left\langle x_K^{\alpha, \beta}(u_K), x_K^{\alpha, \beta}(v_K) \right\rangle_e + \left\langle x_{K'}^{\alpha, \beta}(u_{K'}), x_{K'}^{\alpha, \beta}(v_{K'}) \right\rangle_e \\ & - \left\langle x_{K'}^{\alpha, \beta}(u_{K'}), y_K^{\alpha, \beta}(v_K) \right\rangle_e - \left\langle x_K^{\alpha, \beta}(u_K), y_{K'}^{\alpha, \beta}(v_{K'}) \right\rangle_e. \end{aligned} \quad (29)$$

A direct expansion gives

$$\mathcal{U}_e^{\rightarrow} = G_e^{\alpha, \beta} - i\rho_{\alpha, \beta} C_e. \quad (30)$$

Consequently, whenever $4\alpha\beta = 1$, so that $\rho_{\alpha, \beta} = 1$,

$$A_{P,e} = i\mathcal{U}_e^{\rightarrow}. \quad (31)$$

Thus the complete PWDG interior edge form is an ordinary one-way UWVF on the entire curve $4\alpha\beta = 1$. The physical-impedance condition $\eta_{\alpha, \beta} = k$ intersects this curve only at $\alpha = \beta = 1/2$.

For arbitrary positive α, β , introduce also the reverse-orientation one-way form

$$\begin{aligned} \mathcal{U}_e^{\leftarrow}(u, v) = & \left\langle y_K^{\alpha, \beta}(u_K), y_K^{\alpha, \beta}(v_K) \right\rangle_e + \left\langle y_{K'}^{\alpha, \beta}(u_{K'}), y_{K'}^{\alpha, \beta}(v_{K'}) \right\rangle_e \\ & - \left\langle y_{K'}^{\alpha, \beta}(u_{K'}), x_K^{\alpha, \beta}(v_K) \right\rangle_e - \left\langle y_K^{\alpha, \beta}(u_K), x_{K'}^{\alpha, \beta}(v_{K'}) \right\rangle_e. \end{aligned} \quad (32)$$

It is obtained from the forward form by reversing the sign of the normal-derivative channel in the impedance variables. Therefore the same expansion that gives (30) yields

$$\mathcal{U}_e^{\leftarrow} = G_e^{\alpha, \beta} + i\rho_{\alpha, \beta} C_e. \quad (33)$$

This is an orientation reversal, not the Hilbert-space adjoint of $\mathcal{U}_e^{\rightarrow}$; edgewise C_e is not Hermitian in general.

Set $\rho = 2\sqrt{\alpha\beta}$ and define the two-orientation form

$$\mathcal{U}_e^{\alpha, \beta} = \frac{\rho + 1}{2\rho} \mathcal{U}_e^{\rightarrow} + \frac{\rho - 1}{2\rho} \mathcal{U}_e^{\leftarrow}. \quad (34)$$

Using (30) and (33),

$$\mathcal{U}_e^{\alpha, \beta} = G_e^{\alpha, \beta} - iC_e, \quad A_{P,e} = i\mathcal{U}_e^{\alpha, \beta}. \quad (35)$$

No extra tuning parameter has been introduced: both coefficients in (34) are fixed by the original PWDG penalties. They sum to one, so (34) is an affine combination. It is a convex combination only when $\rho \geq 1$; for $\rho < 1$ the coefficient of the reverse orientation is negative.

The boundary parameter δ has the analogous Riesz interpretation

$$\frac{\delta}{k} \|\partial_n v\|_\Gamma^2 + (1 - \delta) k \|v\|_\Gamma^2,$$

but the standard physical one-way UWVF boundary term occurs at $\delta = 1/2$, exactly as shown in Theorem 1. For the classical PWDG–UWVF equivalence, therefore, all three parameters are one-half.

6. Riesz normal forms

Let $\{\Phi_j\}_{j=1}^N$ be a basis of V_h . Define the PWDG matrix A_P and the TDG Gram matrix G_P by

$$(A_P)_{j\ell} = A_h(\Phi_\ell, \Phi_j), \quad (G_P)_{j\ell} = (\Phi_\ell, \Phi_j)_{\text{TDG}}.$$

Thus, for $v = \sum_j c_j \Phi_j$,

$$c^* A_P c = A_h(v, v), \quad c^* G_P c = \|v\|_{\text{TDG}}^2.$$

Theorem 2 (PWDG Riesz normal form). *The TDG Gram matrix G_P is positive definite on V_h . Moreover,*

$$G_P = \frac{A_P - A_P^*}{2i}. \quad (36)$$

With $B_P = G_P^{-1/2}$,

$$B_P^* A_P B_P = K + iI, \quad K = K^*. \quad (37)$$

Hence the Riesz-normalized PWDG matrix is normal and its spectrum lies on $\text{Im } z = 1$.

Proof. First, G_P is positive definite. If $\|v\|_{\text{TDG}} = 0$, then both interior jumps vanish, so the broken Trefftz field and its normal derivative match across every interior edge. Hence v is a global Helmholtz solution. The two positive boundary terms in (9) give $v = 0$ and $\partial_n v = 0$ on Γ ; uniqueness for the global Cauchy problem then yields $v \equiv 0$. Thus $\|\cdot\|_{\text{TDG}}$ is a norm on V_h , and its Gram matrix is positive definite in exact arithmetic.

We next derive (2) directly. On an interior edge $e = \partial K^+ \cap \partial K^-$, the consistency contribution is

$$C_e(v, v) = \int_e \left(\{\{v\}\} [\nabla \bar{v}]_N - \{\{\nabla v\}\} \cdot [\bar{v}]_N \right) ds.$$

Expanding the averages and jumps gives

$$\begin{aligned} 2C_e(v, v) &= \int_e \left(v^+ \overline{\partial_{n^+} v^+} - \partial_{n^+} v^+ \bar{v}^+ + v^- \overline{\partial_{n^-} v^-} - \partial_{n^-} v^- \bar{v}^- \right) ds \\ &\quad + 2 \text{Re} \int_e \left(v^+ \overline{\partial_{n^-} v^-} + v^- \overline{\partial_{n^+} v^+} \right) ds. \end{aligned}$$

The second line is real, and therefore

$$\operatorname{Im} C_e(v, v) = \operatorname{Im} \int_e \left(v^+ \overline{\partial_{n^+} v^+} + v^- \overline{\partial_{n^-} v^-} \right) ds. \quad (38)$$

On the physical boundary, the consistency part of (7) has imaginary part $\operatorname{Im}(v \overline{\partial_n v})$. Since v is Trefftz on every element, Green's identity gives

$$\sum_{K \in \mathcal{T}_h} \int_{\partial K} v \overline{\partial_{n_K} v} ds = \sum_{K \in \mathcal{T}_h} \int_K (|\nabla v|^2 - k^2 |v|^2) dx \in \mathbb{R}.$$

Splitting the element boundaries into interior and physical edges and using (38), the imaginary parts of all consistency terms cancel. The remaining imaginary terms are exactly

$$\begin{aligned} \operatorname{Im} A_h(v, v) &= \alpha k \|\llbracket v \rrbracket_N\|_{\mathcal{F}_h^I}^2 + \frac{\beta}{k} \|\llbracket \nabla v \rrbracket_N\|_{\mathcal{F}_h^I}^2 \\ &\quad + (1 - \delta) k \|v\|_{\Gamma}^2 + \frac{\delta}{k} \|\partial_n v\|_{\Gamma}^2 = \|v\|_{\text{TDG}}^2. \end{aligned}$$

For every coefficient vector c , this says

$$c^* \left(\frac{A_P - A_P^*}{2i} - G_P \right) c = 0.$$

The matrix in parentheses is Hermitian, so the complex polarization identity proves (36). Writing $H = (A_P + A_P^*)/2$, one has $A_P = H + iG_P$, hence

$$B_P^* A_P B_P = B_P^* H B_P + iI,$$

and $K = B_P^* H B_P$ is Hermitian. $\square$

If $Hx = \mu G_P x$, then μ is invariant under every nonsingular change of Trefftz basis. The normalized eigenvalues are $\mu + i$, and because (37) is normal its singular values are

$$\sigma = \sqrt{1 + \mu^2}. \quad (39)$$

Thus the pencil (H, G_P) describes the intrinsic Riesz spectrum, whereas $\kappa_2(A_P)$ also contains basis conditioning. If tiny eigenvalues of G_P are truncated, the same normal form holds on the retained positive-definite range.

We now specialize the UWVF matrix relations to the classical case $\alpha = \beta = \delta = 1/2$, which is the only case used in the numerical experiment. Let A_U^w denote the one-way UWVF matrix in the weighted trace basis induced by $x_K^{1/2, 1/2} = (2k)^{-1/2} x_K$. By (31) and the trace-space identification of Section 3,

$$A_U^w = -iA_P, \quad G_U^w := A_U^w + (A_U^w)^* = 2G_P. \quad (40)$$

Hence, with $B_U^w = (G_U^w)^{-1/2}$,

$$(B_U^w)^* A_U^w B_U^w = \frac{1}{2} I + iK_U, \quad K_U = K_U^*, \quad K_U = -\frac{1}{2} K. \quad (41)$$

Thus the two normalized matrices are not independent geometries: the UWVF normal form is exactly $-i/2$ times the PWDG normal form.

At the standard one-half parameters, the physical impedance variables (10) are $\sqrt{2k}$ times the weighted traces. Therefore

$$A_U^{\text{std}} = -2ikA_P, \quad G_U^{\text{std}} := A_U^{\text{std}} + (A_U^{\text{std}})^* = 4kG_P. \quad (42)$$

After Riesz normalization, the factor $2k$ disappears. The standard UWVF therefore has the same normalized form (41), separating a trace-normalization convention from the intrinsic operator geometry.

7. Classical UWVF scaling and conditioning

The original discrete UWVF literature writes the algebraic problem in the form

$$(D - C)X = b, \quad (I - D^{-1}C)X = D^{-1}b, \quad (43)$$

where $D = D^* > 0$ is block diagonal and is assembled from the local impedance-trace inner products, while C contains the interelement and boundary coupling; see [5, 6]. The second form is usually regarded as a natural block left-preconditioning because D^{-1} is local. The UWVF literature also recognized that large condition numbers of the element blocks of D are a principal source of numerical instability and used local basis reduction to control them [5, 6].

There are, however, two distinct matrix representations hidden in (43). Define

$$A_L := I - D^{-1}C, \quad A_T := D^{-1/2}(D - C)D^{-1/2} = I - D^{-1/2}CD^{-1/2}. \quad (44)$$

Then

$$A_L = D^{-1/2}A_TD^{1/2}. \quad (45)$$

Thus A_L and A_T are similar and have exactly the same eigenvalues, but generally not the same Euclidean singular values. The distinction is important because A_T , rather than A_L , is the representation of the UWVF coupling in coordinates orthonormal with respect to the trace Gramian D . Indeed, for $\hat{X} = D^{1/2}X$,

$$X^*DX = \|\hat{X}\|_2^2,$$

and the action of $D^{-1}C$ in the D -norm is represented in Euclidean coordinates by $D^{-1/2}CD^{-1/2}$.

Proposition 2 (One-sided versus symmetric trace scaling). *Let $D = D^* > 0$, $A_U = D - C$, and define A_L, A_T by (44). Then*

$$\kappa_2(A_L) \leq \kappa_2(D) \kappa_2(A_T), \quad \kappa_2(A_T) \leq \kappa_2(D) \kappa_2(A_L). \quad (46)$$

If $\|D^{-1/2}CD^{-1/2}\|_2 \leq q < 1$, then

$$1 - q \leq \sigma_j(A_T) \leq 1 + q, \quad \kappa_2(A_T) \leq \frac{1 + q}{1 - q}. \quad (47)$$

Proof. Equation (45) and submultiplicativity give

$$\|A_L\|_2 \leq \|D^{-1/2}\|_2 \|A_T\|_2 \|D^{1/2}\|_2,$$

with the same estimate for the inverse; multiplying the two inequalities gives the first bound in (46), and the reverse similarity gives the second. Writing $U_D = D^{-1/2}CD^{-1/2}$, the reverse triangle inequality yields

$$(1 - q)\|z\|_2 \leq \|(I - U_D)z\|_2 \leq (1 + q)\|z\|_2,$$

which proves (47). $\square$

The trace normalization A_T should not be confused with the graph–Riesz normalization $(G_U^{\text{std}})^{-1/2}A_U^{\text{std}}(G_U^{\text{std}})^{-1/2}$ derived above. The matrix D measures the norm of the UWVF trace unknown itself, whereas $G_U^{\text{std}} = A_U^{\text{std}} + (A_U^{\text{std}})^*$ measures the global transmission-defect norm induced by the Trefftz-DG stability identity. The former removes coordinate distortion already at the level of the trace basis; the latter additionally exposes the intrinsic graph geometry of the coupled discrete problem. Section 8 compares all three levels numerically.

8. Numerical verification

We use a shape-regular unstructured Delaunay triangulation of the unit square with eight triangles, $k = 4$, and p equispaced centroided plane waves per element. The mesh has the four corners, side vertices $(0.48, 0)$ and $(0.55, 1)$, and interior vertices $(0.3467, 0.5113)$ and $(0.6548, 0.4707)$; its minimum angle is 33.1° .

No numerical quadrature is used. Every plane-wave edge product is integrated analytically from

$$\int_e e^{ik\xi \cdot x} ds = |e| e^{ik\xi \cdot (x_a + x_b)/2} \text{sinc}\left(\frac{k\xi \cdot (x_b - x_a)}{2}\right), \quad \text{sinc } t = \frac{\sin t}{t}. \quad (48)$$

PWDG and UWVF are assembled independently at $\alpha = \beta = \delta = 1/2$. The UWVF assembly uses the impedance variables (10). By Theorem 1, its unscaled matrix satisfies $A_U^{\text{std}} = -2ikA_P$, so scalar multiplication leaves the raw condition number unchanged. Likewise $G_U^{\text{std}} = 4kG_P$, so the Riesz-normalized condition numbers coincide.

Table 1: Eight-triangle unstructured mesh with analytic edge integration. The UWVF condition numbers are omitted because they are forced to be identical by the exact scalar relations (42).

p	$N = 8p$	$\kappa_2(A_P)$	$\kappa_2(\hat{A}_P)$
5	40	5.54×10^1	2.134
9	72	2.43×10^5	3.184
13	104	3.21×10^9	4.924

To isolate the effect of the classical left scaling, we also assemble the UWVF directly as $A_U^w = D - C$ in the weighted impedance basis and compare the matrices in (44). The

same analytic edge integrals are used; no matrix in the table is inferred from the PWDG relation.

Table 2: Conditioning of the two classical UWVF reformulations and the graph–Riesz form on the same eight-triangle mesh. Here $A_L = I - D^{-1}C$, $A_T = D^{-1/2}(D - C)D^{-1/2}$, and $\hat{A}_U = G_U^{-1/2}(D - C)G_U^{-1/2}$, with $G_U = (D - C) + (D - C)^*$.

p	$\kappa_2(D)$	$\kappa_2(A_L)$	$\kappa_2(A_T)$	$\kappa_2(\hat{A}_U)$
5	3.79×10^1	1.41×10^1	7.403	2.134
9	1.24×10^5	8.96×10^2	7.499	3.184
13	1.64×10^9	3.80×10^6	9.615	4.924

The one-sided form substantially reduces the condition number of the raw UWVF matrix, but it still inherits severe Euclidean distortion from the trace Gramian: between $p = 5$ and $p = 13$, $\kappa_2(D)$ grows by more than seven orders of magnitude and $\kappa_2(A_L)$ grows from about 14 to 3.8×10^6 . In contrast, the symmetric trace-Riesz matrix remains below 10 throughout these rows. The graph–Riesz normalization is smaller still, because it removes not only the trace-coordinate metric but also uses the global transmission-defect metric generated by the UWVF/PWDG stability identity.

Here

$$\hat{A}_P = G_P^{-1/2} A_P G_P^{-1/2}, \quad \hat{A}_U = (G_U^{\text{std}})^{-1/2} A_U^{\text{std}} (G_U^{\text{std}})^{-1/2}.$$

The independent assemblies satisfy

$$\frac{\|A_U^{\text{std}} + 2ikA_P\|_F}{\|A_U^{\text{std}}\|_F} \leq 2.1 \times 10^{-16}, \quad \frac{\|G_U^{\text{std}} - 4kG_P\|_F}{\|G_U^{\text{std}}\|_F} \leq 2.3 \times 10^{-16} \quad (49)$$

over the three displayed rows. These independently assembled residuals, rather than the equal condition numbers, are the numerical verification of the matrix equivalence. The full-impedance UWVF and the PWDG matrix are recovered from one another to roundoff on a multi-element unstructured mesh, without quadrature.

The table also illustrates the distinction between raw and Riesz conditioning in a regime where the TDG Gram matrix is still numerically resolvable in double precision. Increasing the local plane-wave count from $p = 5$ to $p = 13$ increases the raw condition number by almost eight orders of magnitude, while the Riesz-normalized value grows only from about 2.13 to 4.92. At $p = 17$, the smallest eigenvalues of G_P lie at the double-precision noise floor and the untruncated inverse square root is no longer a reliable numerical operation; that row is therefore not reported. In practice one should either use higher precision or state an explicit spectral truncation threshold and retained dimension.

9. Discussion

The direct formula (14) is the simplest way to state the classical equivalence: at $\alpha = \beta = \delta = 1/2$, PWDG is the UWVF written in broken Trefftz fields. The left-hand side of the UWVF uses the outgoing and incoming impedance variables, while the equivalent PWDG description uses averages and the actual Dirichlet and normal-flux jumps. There is no additional skeleton unknown once the impedance traces are pulled back by the local Trefftz solutions.

Proposition 1 explains why the two interior penalties have the values they do. The two oriented impedance transmission conditions contain the same information as the pair

$$[[u]]_N, \quad [[\nabla u]]_N.$$

Averaging the two oriented impedance products cancels the mixed value–flux terms and leaves precisely the two positive PWDG jump products. At the physical impedance, the normalization needed to produce these two channels is exactly $\alpha = \beta = 1/2$.

For general positive α and β , the same algebra gives a useful interpretation of the PWDG flux family. The ratio α/β chooses the impedance $k\sqrt{\alpha/\beta}$. The product $\alpha\beta$ controls the scale and the affine weighting of the forward and reverse orientations. On $4\alpha\beta = 1$ the reverse coefficient vanishes and the representation reduces to a one-way UWVF; away from that curve the exact representation is the fixed two-orientation combination (34). For $4\alpha\beta < 1$, the reverse coefficient is negative, so the combination should not be called a mixture or convex average.

The classical UWVF scaling of Section 7 sharpens this distinction. The one-sided matrix $I - D^{-1}C$ is isospectral to the symmetrically trace-normalized matrix but can have very different singular values when the local trace Gramian D is ill-conditioned. Thus the classical D^{-1} scaling captures the correct generalized spectrum without, by itself, removing Euclidean coordinate distortion.

The Riesz identities provide the corresponding operator interpretation. In broken-field coordinates,

$$A_P = H + iG_P,$$

with G_P the TDG jump-and-boundary Gram matrix. Riesz coordinates turn the dissipative part into the identity and give $K + iI$. In UWVF coordinates the corresponding trace Riesz map gives $\frac{1}{2}I + iK_U$, which is exactly the fixed scalar $-i/2$ times the PWDG normal form. The different raw matrices therefore represent the same edge-coupling problem in two coordinate systems rather than two independent operator geometries. This viewpoint is closely related to operator preconditioning [4] and to local UWVF boundary orthonormalization [5], but here the Riesz map is obtained directly from the PWDG coercivity identity.

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